

Shohini Rhea Sarkar

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EDUCATION

University of Maryland

College Park, MD

B.S. Computer Science: Machine Learning — GPA: 3.94

Expected May 2027

- **Coursework:** Algorithms, Intro to AI, Natural Language Processing, Data Science, Object Oriented Programming, Computer Systems, Linear Algebra, Discrete Math, Probability & Statistics, Multivariable Calculus

EXPERIENCE

Incoming Software Development Engineer Intern

Summer 2026

Amazon Web Services (AWS)

San Diego, CA

- Selected for SDE internship at AWS; will work on security

Research Intern

May 2025 – Present

University of Maryland / Army Research Laboratory

College Park, MD

- Designed deep learning peak detection pipelines to extract neuronal firing events from calcium imaging data
- Developed neural connectivity pipelines using logistic regression to quantify neuron–neuron influence
- Processed over **2TB** of two-photon imaging data using custom scripts to produce analysis-ready formats

Machine Learning Technical Lead

Sept. 2024 – May 2025

Amazon Project Leo, UMD App Dev

College Park, MD

- Promoted from ML Engineer to Technical Lead, directing a 10+ person team working on satellite ML systems
- Designed satellite state transition classifiers to achieve **97%** accuracy across multi-class prediction tasks
- Developed a clutter height prediction pipeline using feature engineering to improve accuracy from **44% to 76%**

AI Research Intern

June 2023 – August 2023

UChicago Data Science Institute / Fermi National Laboratory

Chicago, IL

- Developed RL agents using policy optimization to automate telescope positioning for astronomical event detection
- Tuned exploration strategies and reward functions in resource-constrained environments; presented findings at the UChicago DSI Symposium

Research Intern

June 2022 – August 2022

Lawrence Berkeley National Laboratory

Berkeley, CA

- Implemented the Variational Quantum Eigensolver using Qiskit to compute molecular ground state energies across multiple quantum backends
- Contributed to readout error mitigation algorithms to improve quantum error correction accuracy

PROJECTS

RL for Insulin Pump Optimization | Python, OpenAI Gym, PyTorch

2022–2023

- Built policy-based RL agents using reward shaping to stabilize glucose levels in simulated environments; published at **MIT URC '23, 1st place** at Alameda County Science Fair

AI for Drug Discovery | Python, TensorFlow, ATOM Pipeline

2021–2022

- Integrated deep learning improvements into the ATOM drug candidate prioritization pipeline using loss function tuning to improve precision; published at **MIT URTC '22**

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL

Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, CaImAn

Tools: Git, Google Colab, Jupyter Notebooks, ML Pipelines

HONORS

Presidential Scholar Candidate (U.S. Dept. of Education, Mar. 2024): Selected as one of 6,000 most distinguished graduating seniors in the nation

SAT Perfect Scorer